Ethers

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1 Ether Synthesis

Ethers are a class of organic compounds characterized by an oxygen atom bonded to two alkyl or aryl groups. Their general formula is R—O—R', where R and R' can be the same (symmetrical ether) or different (unsymmetrical ether). Here is a simple example of an ether: dimethyl ether.

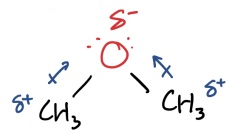

Figure 1: Dimethyl Ether Structure

In the context of nomenclature, the larger group is defined as the stem and other half is called an alkoxy substituent. Take a look at the following compounds.

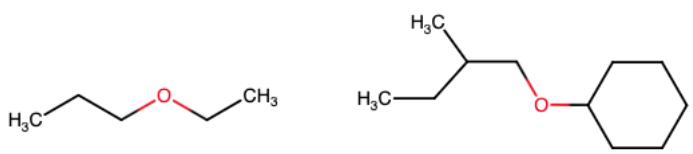

Figure 2: Ether Nomenclature Examples

The respective names of each compound are the following

1-Ethoxypropane (2-Methylbutoxy)cyclohexane

Now let's get into ether synthesis. The primary way chemists make **asymetrical** ethers is through the **WILLIAMSON ETHER SYNTHESIS** method. This method uses an alkoxide $R'O^-$ as its Nu, thus works best when attacking unhindered primary $R-X$ because hindered, secondary, or tertiary $R-X$ will switch over to the E2 mechanism and result in an unintended product.

$$R_{prim}-X + R'O^- \xrightarrow{SN2} R-O-R'$$

It's important to note here that the if the desired product is asymetrical, the choice of cooresponding R groups paired with the halogen and oxygen atom is important because a tertiary or secondary haloalkane pared with a primary alkoxide will lead to E2 whereas the opposite combination will undergo SN2 (*the desired method*).

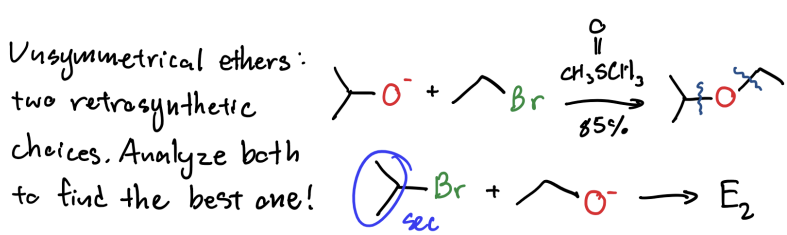

Figure 3: Williamson Ether Synthesis Retrosynthesis

In order to create ethers for $R_{sec/tert}X$, we use the protonated version of alkoxides (a.k.a. alcohol) to make the Nu

less basic and consequently undergo SN1 or E1 rather than E2. This is simply the **ALCOHOLYSIS OF $R_{sec/tert}X$.**

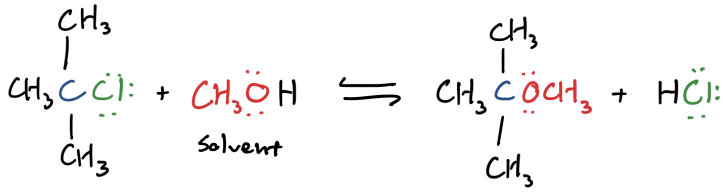

Figure 4: Alcoholysis of Alkyl Halides

While potential E1 reactions are still an issue that results in unintended products, we can minimize this by carrying out the reaction in lower temperatures.

So far, we've looked at **intermolecular** ether synthesis reactions. Now we introduce **INTRAMOLECULAR WILLIAMSON ETHER SYNTHESIS** (a.k.a. **cyclic ether formation**). These form in the same way as the intermolecular equivalent has done.

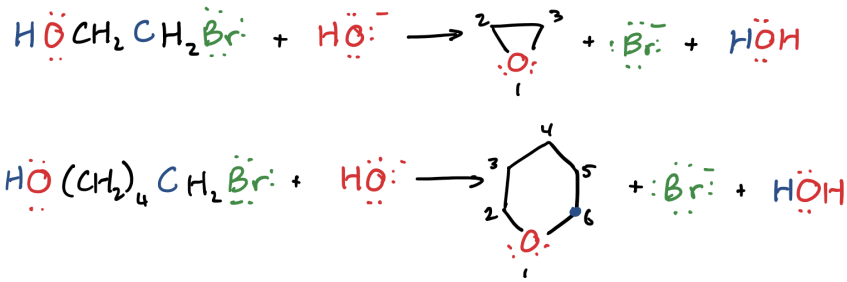

Figure 5: Intramolecular Williamson Ether Synthesis

Intramolecular are particularly favorable over intermolecular because [1] it does not lower entropy and [2] highly diluted concentrations of the relevant alcohol do not effect it as much because the reaction is occurring with itself.

Mechanistically speaking, this reaction occurs via a backside attack with inversion, prepared by the rearrangement to a less stable rotamer if needed.

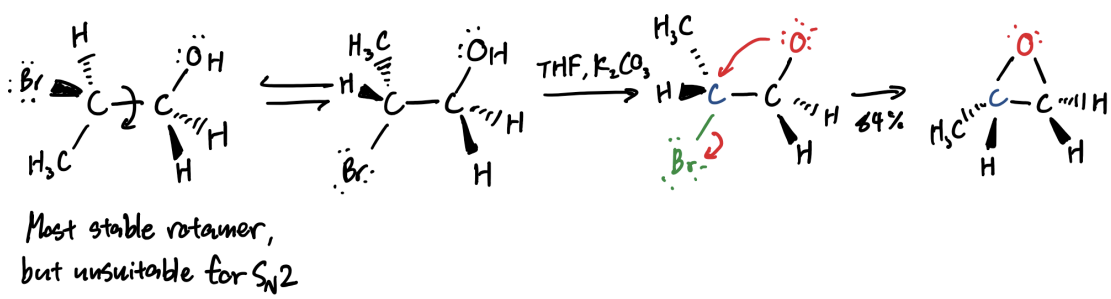

Figure 6: Backside Attack and Inversion in Cyclic Ether Formation

It's important to keep in mind that the relative rates of ring closure depend on the trade off between ΔH (*ring strain*) and ΔS (*lining up*) in the various transition states. Here is the ranking of ether ring formation speeds by ring size.

$$3 > 5 \geq 6 > 4 \geq 7 > 8$$

Notice how it does not nessecarily coorespond directly to ring size. For instance, a 3 membered ring is the fastest because of the proximity of the bonding atoms. A 4 membered ring is slow because of the distance between the atoms. However, a 6 membered ring is again fast because its arrangement decreases the distance between the atoms relative to the 4 membered ring. The nuances and detailed understanding behind this ordering is unnessecary to know for the purposes of this course. It's sufficient to just realize that it's due to the trade offs between ΔH and ΔS of the various transitions states across ring sizes.

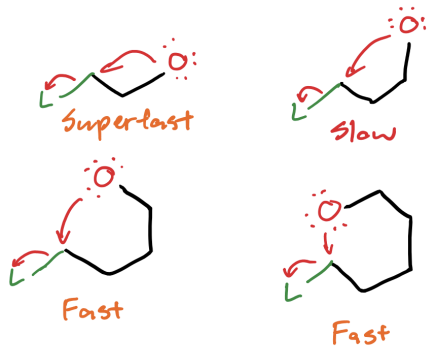

Figure 7: Relative Rates of Ring Closure

Ethers can also be formed by **alcohols and H^+ through SN2 and/or SN1** depending on the R groups in question. This is specifically done to form **symmetrical** ethers because an alcohol mixture would produce all possible combinations rather than one specific product.

$$2\,\mathrm{R-OH} \xrightarrow[-\mathrm{H_2O}]{\mathrm{H_2SO_4}} \mathrm{R-O-R}$$

For $\mathrm{R_{prim}-OH}$, the mechanism works via SN2 because there is minimal steric hinderance.

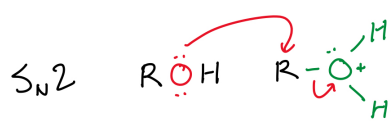

Figure 8: Symmetrical Ether Formation via SN2

When dealing with $\mathrm{R_{sec/tert}-OH}$, the mechanism shifts to SN1.

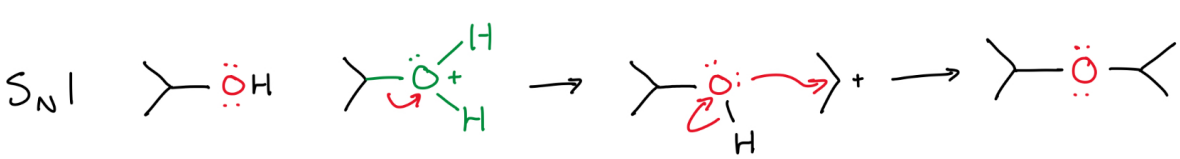

Figure 9: Symmetrical Ether Formation via SN1

However, a case where a combination of alcohols in an acidic solution preferentially form an asymetrical ether is an **unhindered ROH and tetrabutyl scenario**.

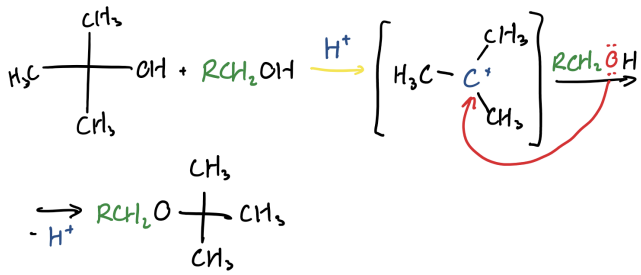

Figure 10: Asymmetrical Ether Formation with t-Butyl Group

This reaction holds a preference for the ROH attacking the tetrabutyl cation via SN1 because the t-Bu cation forms extremely quick due to the high stablity as a result of hyperconjugation. It is also less likely for the unhindered ROH to form a cation because it has lower levels of potential hyperconjugation. And while the t-Bu cation could be attacked by another bulky t-Bu alcohol molecule, steric hinderance makes this combination less likely to happen compared to the smaller, less sterically hindered ROH.

In the laboratory, tetrabutyl is used for **"protection"** of other alcohols by forming the t-butyl ethers we have just learned about by removing the acidic H.

2 Ether Reactions

Let's go through a collection of different ether reactions.

(1) $R_{prim} - O - R_{prim}$ are quite unreactive. They are stable to bases, RLi, RMgBr, and dilute aqueous H^+. However they are reactive under H^+X^- exposure

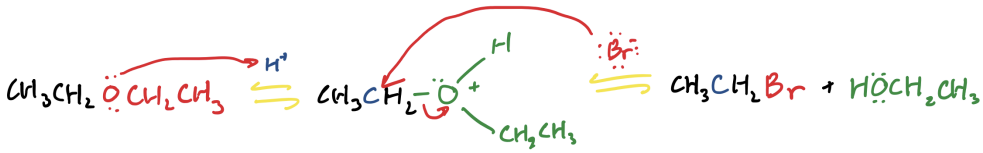

Figure 11: Ether Cleavage with Hydrohalic Acids

(2) $R_{prim} - O - R_{sec}$ mixed ethers react either through SN1 or SN2 depending on the Nu strength (*under acidic conditions*). When paired with a **good Nu**, the Nu attacks the R_{prim} side carbon of the ether through SN2 to avoid the steric hinderance of the R_{sec} side.

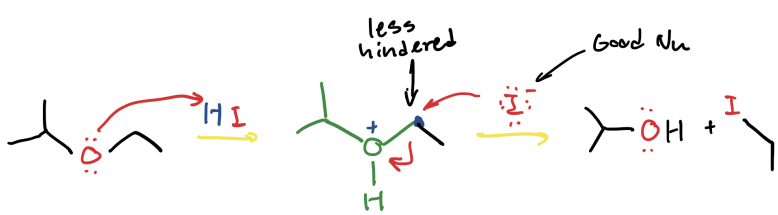

Figure 12: Mixed Ether Reaction with Good Nucleophile (SN2)

When paired with a **poor Nu**, the Nu undergoes an SN1 reaction and attacks the the carbocation that forms on the R_{sec} because a primary carbocation is too unstable to form in the first place.

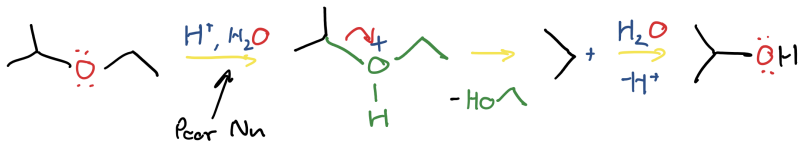

Figure 13: Mixed Ether Reaction with Poor Nucleophile (SN1)

(3) Tetrabutyl ether hydrolysis is the reverse of t-Bu ether formation we learned previously in **2.3.1 Ether Synthesis**. This reverse process is the "deprotection" of alcohol ROH by t-Bu cation via acid. The resulting t-Bu cation, when exposed to heat undergoes E1 to exaporate into an alkene.

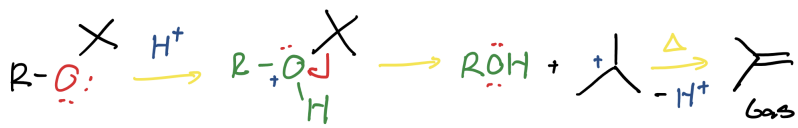

Figure 14: Hydrolysis and Deprotection of t-Butyl Ethers

(4) Ether ring openings release ring strain by a Nu^- attacking directly one of the two C adjacent to the O. Keep in mind that this process is regioselective, meaning the SN2 occurs at the less hindered site of the two carbons.

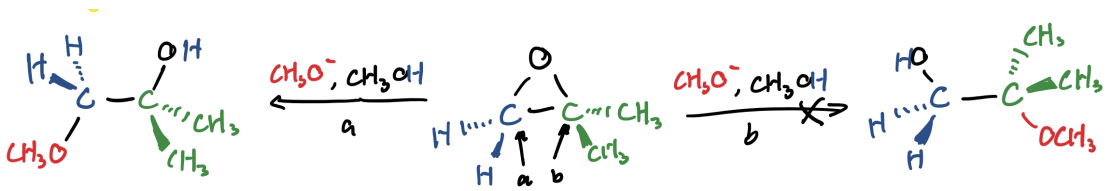

Figure 15: Regioselective Ether Ring Opening

Many Nu^- will work. In this example, we show that ether ring openings are also regioselective and **stereocontrolled**.

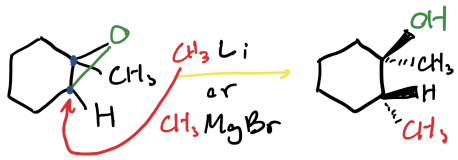

Figure 16: Stereocontrolled Ether Ring Opening

With **neutral Nu**, we need acidic conditions to activate the ether to a nucleophilic attack. This reaction WILL NOT OCCUR without H^+ present.

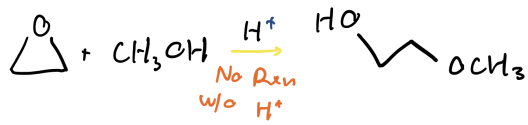

Figure 17: Acid-Activated Ether Ring Opening with Neutral Nucleophile

One important aspect difference about this neutral Nu mechanism is that in asymmetrical ethers, the Nu is more regioselective to the more hindered side because the more substituted C is better at stabilizing a partial positive charge (δ+) through hyperconjugation and inductive effects, thus the C-O bond on that side starts to "stretch" and weaken significantly. This is sort of an SN1 and SN2 hybrid scenario (SN2 because the Nu attack and L group departure happens simultaneously; SN1 because the partial positive is similar to a cation).

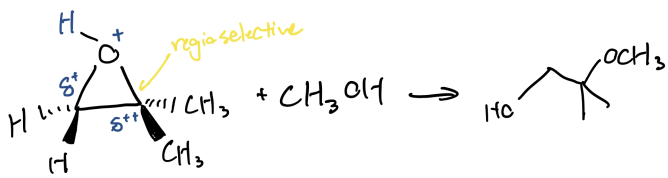

Figure 18: Regioselectivity in Acid-Catalyzed Ring Opening of Asymmetrical Ethers

3 Sulfur Analogs: *A Brief Lesson*

Sulfur (S) is directly below oxygen (O) in the periodic table, thus is behaves very similarly to how oxygen would in the context of bonding.

The sulfur analogs of ROH are called alkane**thiols**.

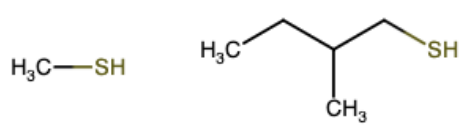

Figure 19: sulfer nomenclature 1

The respective names of the above compounds are

methanethiol 2-methyl-1-butanthiol

As substituients, sulfurs are called either mercapto (–SH) or alkylthio (–SR) groups. Keep in mind that the –OH has priority over the –SH when both are present in the same molecule.

Figure 20: sulfer nomenclature 2

The respective names of the above compounds are

2-mercaptoethan-1-ol methylthioethane

Moving on to characteristics, thiols are **more acidic than ROH** because the RS – H bond is weaker (*the molecular orbitals are at a higher level of energy*). This results in **less H-bonding than ROH** because the EN difference isn't as high as that of ROH. Consequently, thiols are less polar. *H_2S is a gas at 25°C (boiling point -60°C).*

However, although RS^- is less basic than RO^-, RS^- is still a better Nu because it is more **polarizable**. *Mind that polarizable is different from being polar.* Polarizability describes how easily the e^- cloud can be distorted by an outside force, a concept covered in *2.1.1 Nucleophilic Substitution*. Generally larger elements like S (over O) are more polarizable because the size of its e^- cloud results in greater "moldability" compared to the smaller e^- cloud of O. Because S isn't held in a tight, rigid sphere by the nucleus, it can stretch and distort toward the electrophile, making RS^- a better Nu.

This better nucleophility (in comparison to RO^-) eliminates the issue of E2 reactions when paired with $R_{sec}X$ like ROH has.

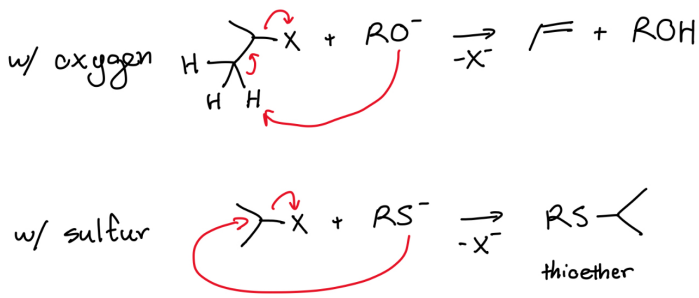

Figure 21: sulfer e2 elimination

Even neutral RSR' undergo SN2 (like NH_3, PR_3) whereas dimethyl ether (CH_3OCH_3) has no reaction. Notice how if the RSR' nucleophilically attacks an RX, the resulting R_3S^+ is susceptible to an attack by a different nucleophile. *The neutral RSR' that was once a good nucleophile has become a good leaving group, like H_2O.* This makes these compounds or **sulfonium salts** (R_3S^+) good **alkylating agents**, which is a compound capable of transferring an alkyl group to another molecule, namely a Nu.

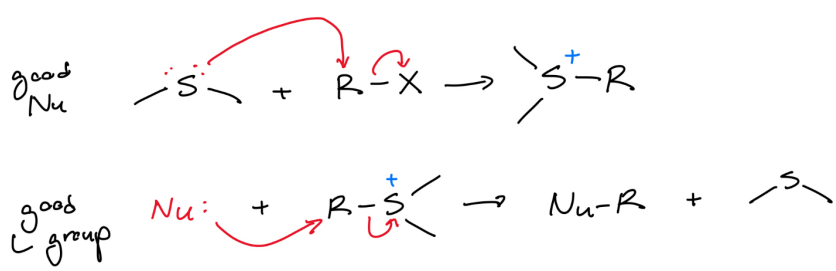

Figure 22: sulfer good nul